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Morphometry of the Lumbar Spine: Anatomical Perspectives Related to Transpedicular Fixation*

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ABSTRACT: **The pedicles of lumbar vertebrae were measured both directly and radiographically to determine the differences between the sexes and the accuracy of radiographic measurement. The lumbar pedicles of cadavera of forty-nine patients — twenty-four men and twenty-five women — who died between the ages of sixty and ninety-eight years were measured directly and on radiographs. The pedicles of lumbar vertebrae from fifty-one patients — twenty-three men and twenty-eight women — between the ages of twenty and fifty years who had low-back problems were measured on radiographs and computerized tomographic scans.**

Comparison revealed that the average transverse and sagittal diameters of the pedicles and the distance from the posterior aspect of the laminar cortex to the anterior aspect of the cortex of the vertebral body along the central axis of the pedicles were 5 to 20 per cent greater in men, but the transverse and sagittal angles of the pedicle did not differ significantly between the sexes.

Measurements on radiographs and computerized tomographic scans of the transverse angles of the pedicles and of the distances from the posterior aspect of the laminar cortex to the anterior aspect of the cortex of the vertebral body from the second to the fifth lumbar vertebra were greater than direct measurements, even without magnification. Direct measurements of the diameters of the transverse and sagittal diameters of the pedicle of the fifth lumbar vertebra, however, were greater than the radiographic measurements.

Transpedicular fixation of the lumbar spine with screws is becoming increasingly popular[62]. Several systems of internal fixation that attach through the pedicles are currently available[18,24-26,33,34,36-40,48-50,53,54]. These systems depend on the ability of screws traversing pedicles — the strongest parts even of osteoporotic vertebrae[26,30,47] — to hold in the vertebral bodies until solid fusion occurs[4,6,10,23,24,35,44,47,56,62].

Procedures that have been developed for the treatment of younger patients who have scoliosis are not well suited for the lumbar spine, for several reasons[26,61,62]. The lumbar spine is normally lordotic, and this lordosis must be preserved. Most patients who have posterolateral arthrodesis and internal fixation of the lumbar spine are older than fifty years[42,61], and many are severely osteoporotic[27,41,42,46]; most instrumentations, therefore, do not hold well[61]. Because of stenosis of the spinal canal, many elderly patients have a wide laminectomy posteriorly for adequate decompression; this may result in instability unless the spine is fused[27,61]. A lateral arthrodesis after a wide midline decompression, especially if articular processes have been sacrificed, often fails to fuse solidly unless some effective internal fixation is added[61].

Accurate anatomical descriptions of the shape and orientation of lumbar pedicles are necessary for the development and use of implantable devices and spinal instrumentation[7,12,14-16,22,26,29,31,43,45,47,52,53,57-60,63]. Several studies have investigated the morphometry of lumbar pedicles with the use of various techniques, such as direct measurement and measurement on plain radiographs or computerized tomographic scans[7,26,47,52,53,57,63].

It is important to distinguish differences in the morphometry of pedicles in men and women and to understand changes in the elderly[2,3,11,13-15,17,21,30]. Data should be obtained from radiographs of spines of cadavera and then compared with direct measurements; these data could then be used to verify the accuracy of the technique that radiologists and surgeons must use for preoperative evaluation[26]. If the dimensions of pedicles change at each vertebral level and vary according to the sex and age of patients, information on these variations might help to prevent failure of fixation and injury to vital structures.

Materials and Methods

The study was divided into three parts: the direct measurement of lumbar pedicles in cadavera, the comparison of measurements on radiographs of lumbar pedicles in cadavera with direct measurements, and the measurement of lumbar pedicles on radiographs and computerized tomographic scans of living young-adult patients.

All diameters and distances were measured with a vernier caliper (precision, ±0.1 millimeter; Inoxyd Helios, Hannover, West Germany), and all measurements of angles were made with a protractor (precision, ± 1 degree) by the

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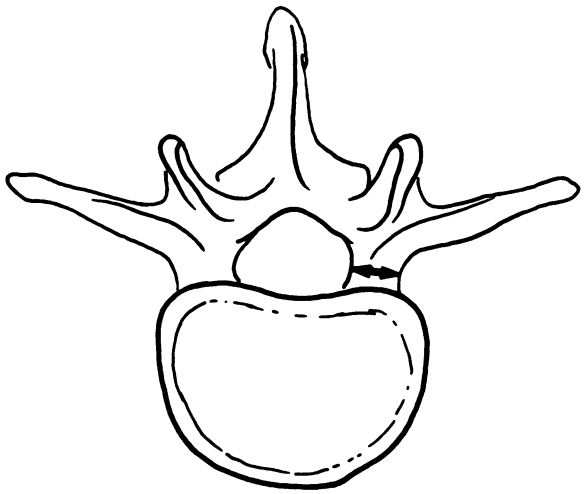


FIG. 1

Drawing of the top of a lumbar vertebra, showing how the transverse diameter (arrows) was measured for both pedicles directly and on radiographs.

same investigator, with an average of three trials for each data-point. The error of measurement was ±0.1 millimeter for each dimension and ±1 degree for each angle.

Lumbosacral spines from forty-nine embalmed cadavera (twenty-four men and twenty-five women), obtained from the Department of Anatomical Sciences, State University of New York at Buffalo School of Medicine, were used for Parts 1 and 2 of the study. The age-range for the men was sixty to ninety years, with a mean of seventy-four years, and the range for the women was sixty to ninety-eight years, with a mean of seventy-seven years.

Part 1: Direct Measurement of Lumbar Pedicles in Cadavera

From the gross specimens, both pedicles at each lumbar level were measured directly to determine the transverse and sagittal diameters; the transverse and sagittal angles of the pedicle, to show the orientation in the transverse and sagittal planes; and the shortest distance from the posterior aspect of the laminar cortex to the anterior aspect of the cortex of the vertebral body in a line that passes through the middle of the pedicle.

The narrowest dimensions in the transverse and sagittal planes were chosen as the transverse and sagittal diameters

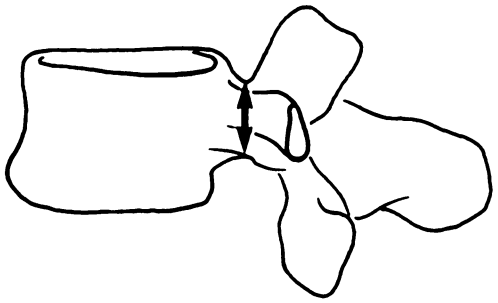


FIG. 2

Side view of a lumbar vertebra, showing how the sagittal diameter (arrows) was measured for both pedicles directly and on radiographs.

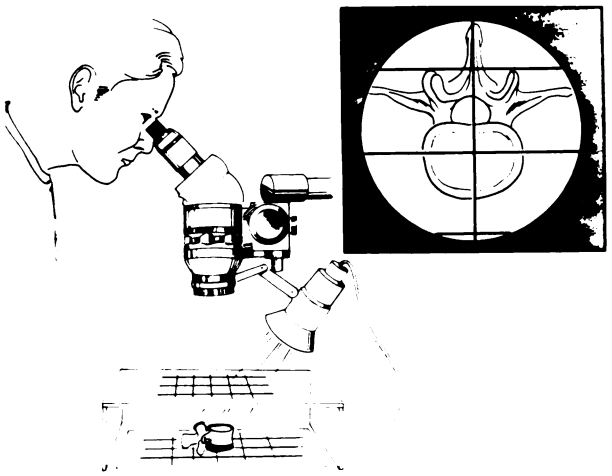


FIG. 3

Drawing showing the use of a dissecting microscope for the orientation and measurement of the transverse and sagittal angles and the distance from the posterior aspect of the laminar cortex to the anterior aspect of the cortex of the vertebral body. The inset shows a vertebra, oriented for measurements in the transverse plane, as seen through the eyepiece of the dissecting microscope.

of the pedicle (Figs. 1 and 2). Consistent orientation of the pedicle in the transverse and sagittal planes was obtained with an instrument consisting of a clear Plexiglas stage with grid lines aligned above and below the specimen (Fig. 3). Mounted above was a dissecting microscope (Model M5-22991; Wild Heerbrug, Zurich, Switzerland) that consisted of an eyepiece, an objective, and a reducing lens (magnification factors, eight, six, and 0.5, respectively). As seen through a single eyepiece, the vertebra was aligned for measurements in the transverse plane with its median plane parallel to the vertical grid-lines on the stage, such that the junction of the inferior end-plate and the posterior aspect

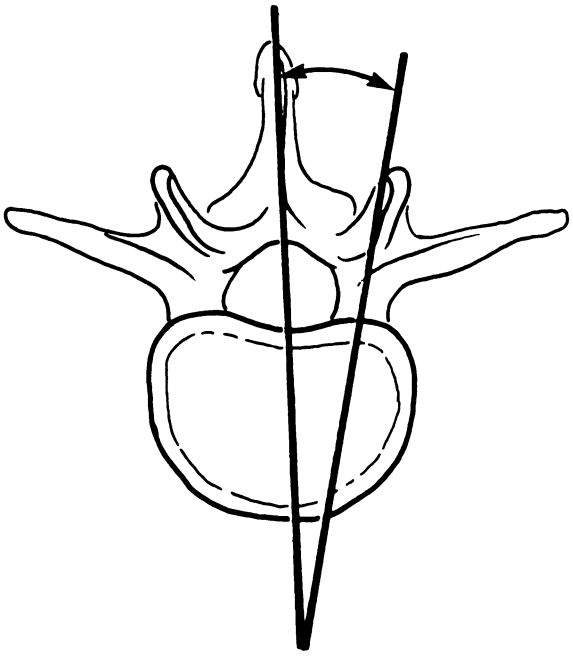


FIG. 4

The transverse angle was measured for both pedicles directly, on radiographs, and on computerized tomographic scans.

of the cortex of the vertebral body could not quite be seen as the superior end-plate of the vertebral body was viewed. Transverse angles were obtained by measuring the angle between a line along the central axis of the pedicle and a line parallel to the median vertebral plane (Fig. 4). Next, the vertebra was aligned on its lateral surface for measurements in the sagittal plane. The superior end-plate of the vertebral body was placed parallel to a horizontal grid-line on the stage, such that the pedicle farthest from the microscope lay, as nearly as possible, superimposed behind the pedicle closest to the microscope, to reproduce the ideal orientation of an ordinary lateral radiograph. The sagittal angle was measured between a line along the central axis of the pedicle and a line parallel to the superior end-plate of the vertebral body (Fig. 5). The vertebra was then reversed, and the procedure was repeated for measurement of the sagittal angle of the opposite pedicle. The distance from the posterior aspect of the laminar cortex to the anterior aspect of the cortex of the vertebral body in a line passing

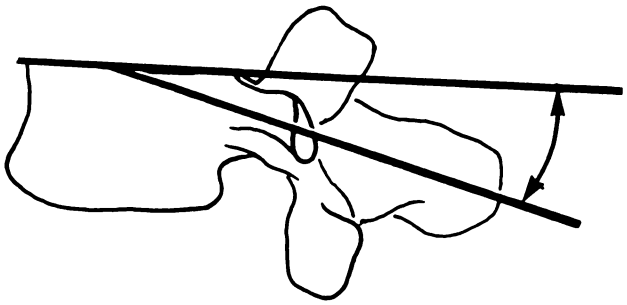


FIG. 5

The sagittal angle was measured for both pedicles directly and on radiographs.

through the middle of the pedicle was measured under the dissecting microscope along the same line that had been used for measurement of the transverse angle (Fig. 6).

Part 2: Comparison of Radiographic Measurements of the Lumbar Pedicles with Direct Measurements

Anteroposterior, lateral, and transverse radiographs were made of twenty-one disarticulated lumbar spines of cadavera (twelve men and nine women), selected at random from the forty-nine specimens that were examined in Part 1. The radiographs were made in the Department of Radiology of the Buffalo General Hospital. The technique was kept constant and was calibrated to produce no measurable magnification, with a target-to-film distance of forty-seven inches (119.4 centimeters) and a focal spot size of 0.6 millimeter. The same measurements that had been obtained from the gross specimens in Part 1 (Figs. 1, 2, 4, 5, and 6) were now obtained from the radiographs: the transverse and sagittal diameters, from the anteroposterior radiographs; the sagittal angles, from the lateral radiographs; and the transverse angle and the shortest distance from the posterior aspect of the laminar cortex to the anterior aspect of the cortex of the vertebral body, from the transverse radiographs. Sagittal angles were described as positive if they were cephalad in direction and as negative if caudad.

Part 3: Measurement of Lumbar Pedicles on Radiographs and Computerized Tomographic Scans of Living Young-Adult Patients

Plain radiographs and computerized tomographic scans of the lumbosacral spine were made for fifty-one patients (twenty-three men and twenty-eight women) who were seen because of low-back problems in the Department of Orthopaedics or Neurosurgery, Buffalo General Hospital. The age-range for the men was twenty to forty-nine years, with a mean of thirty-five years, and for the women it was twenty-one to fifty years, with a mean of thirty-eight years.

The plain radiographs were routine anteroposterior and decubitus lateral radiographs of the lumbosacral spine. The computerized tomographic scans were all made with a Somaton DRH Scanner (Siemens, Erlangen, West Germany) at version G/G1/H; cuts were in contiguous increments of four millimeters. Cuts were made parallel to the planes of the superior end-plates of the vertebral bodies (to compensate for lordosis) and through the middle of the pedicles whenever possible. The gantry of the scanner was adjusted to align it parallel to the end-plates as accurately as possible, but it was not always parallel at the level between the fifth lumbar and first sacral vertebrae. The techniques for making the plain radiographs and the computerized tomographic scans were the same for all patients. When a radiograph showed a gross abnormality of the lumbosacral spine, that patient was excluded from the study.

From the radiographs and computerized tomographic scans, both pedicles at each lumbar level were measured to determine the transverse and sagittal diameters (on the anteroposterior plain radiographs); the sagittal angles (on the lateral radiographs); and the transverse angle and the shortest distance from the posterior aspect of the laminar cortex to the anterior aspect of the cortex of the vertebral body in a line that passes along the central axis of the pedicle (on the computerized tomographic scans).

Magnification on the anteroposterior radiographs was

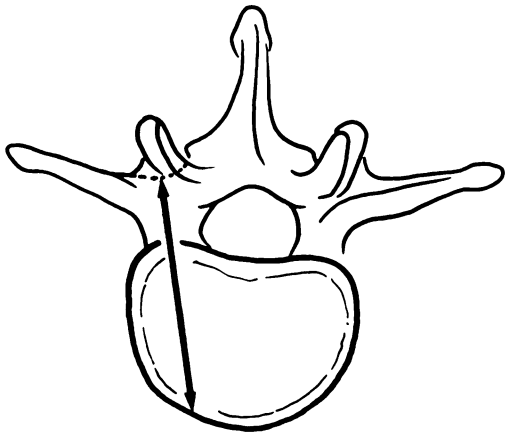


FIG. 6

The distance from the posterior aspect of the laminar cortex to the anterior aspect of the cortex of the vertebral body for both pedicles was measured along a line that passes through the middle of the pedicle. This was done directly, on radiographs, and on computerized tomographic scans.

TABLE I

DIRECT MEASUREMENT OF LUMBAR PEDICLES OF CADAVERA*

Level	Transverse Diameter (mm)		Sagittal Diameter (mm)		Transverse Angle (Degrees)		Sagittal Angle (Degrees)		Posterior-to-Anterior Distance (mm)	
	M	F	M	F	M	F	M	F	M	F
L1	9.5 ± 2.9 (39)	7.7 ± 1.9† (34)	17.0 ± 1.7 (38)	15.3 ± 2.0† (31)	7 ± 2 (37)	5 ± 2† (29)	5 ± 1 (36)	6 ± 2 (26)	45.0 ± 2.7 (31)	42.5 ± 2.6† (24)
L2	9.6 ± 2.2 (40)	7.9 ± 1.9† (33)	16.0 ± 2.2 (39)	15.3 ± 2.0 (32)	7 ± 2 (35)	6 ± 2† (29)	6 ± 2 (35)	5 ± 2 (27)	46.8 ± 3.2 (33)	43.5 ± 3.0† (26)
L3	11.7 ± 2.5 (40)	9.6 ± 2.4† (34)	16.1 ± 2.1 (42)	15.0 ± 1.5† (33)	8 ± 2 (42)	7 ± 2 (33)	6 ± 2 (42)	6 ± 2 (32)	49.0 ± 3.8 (40)	44.8 ± 2.6† (31)
L4	14.7 ± 2.7 (44)	12.5 ± 2.3† (36)	16.4 ± 2.5 (42)	14.9 ± 1.5† (36)	11 ± 4 (44)	10 ± 4 (36)	6 ± 2 (44)	7 ± 2 (35)	50.4 ± 4.6 (41)	45.9 ± 3.0† (34)
L5	21.1 ± 3.4 (47)	18.4 ± 3.6† (39)	17.4 ± 3.4 (47)	16.2 ± 1.9† (39)	17 ± 7 (45)	18 ± 8 (37)	5 ± 2 (38) −8 ± 5‡ (8)	8 ± 3† (33) 9 ± 5‡ (4)	46.2 ± 4.0 (45)	42.3 ± 4.0† (37)

* Data are given as means and standard deviations. The numbers in parentheses are the numbers of examples that were analyzed.
† Statistically significant compared with the value for the men (Student t test, $p \leq 0.05$).
‡ Sagittal angles at the fifth lumbar level occurred in either a cephalad or a caudad direction and are reported separately here.

corrected by making a radiograph of a standard measurement device (Collimator test tool; Radiation Measurements, Middletown, Wisconsin) at the same distance and settings as those used for the patients. A distance of 100 millimeters on the device measured 108 millimeters on the radiograph, so a correction factor of 0.93 (100/108) was used for the transverse and sagittal diameters.

The correction factor for the computerized tomographic scans was also determined; a known distance of 100 millimeters measured only fifty-five millimeters on the scans, so a correction factor of 1.82 (100/55) was used for calculation of the distance from the posterior aspect of the laminar cortex to the anterior aspect of the cortex of the vertebral body. No correction factor was used for measurement of the angles.

Data were recorded in tabular form for each vertebral level. Groups of data in Parts 1 and 3 were analyzed statistically for differences between the sexes at each vertebral level, and pooled data for both sexes in Parts 1 and 3 were analyzed statistically for differences at each vertebral level with the Student t test[9]. Pooled data from all three parts of the study were analyzed for statistical differences at each vertebral level by means of an analysis of variance[9]. Significance for all tests was set at $p \leq 0.05$.

Results

Part 1: Direct Measurement of Lumbar Pedicles in Cadavera (Table I)

The greatest transverse and sagittal diameters were in the fifth lumbar vertebra for both sexes. Statistically significant differences in the diameters were found between the sexes, with consistently smaller diameters at each lumbar level in the specimens from women, with the exception of the sagittal diameters of the second lumbar vertebra.

TABLE II

COMPARISON OF RADIOGRAPHIC MEASUREMENTS OF THE LUMBAR PEDICLES WITH DIRECT MEASUREMENTS*

Level	Transverse Diameter (mm)		Sagittal Diameter (mm)		Transverse Angle (Degrees)		Sagittal Angle (Degrees)		Posterior-to-Anterior Distance (mm)	
	Direct	Radiog.	Direct	Radiog.	Direct.	Radiog.	Direct	Radiog.	Direct	Radiog.
L1	8.2 ± 2.3 (41)	8.2 ± 1.8 (39)	16.4 ± 1.7 (42)	18.2 ± 1.7† (39)	6 ± 2 (42)	6 ± 3 (41)	6 ± 2 (42)	7 ± 3† (38)	43.3 ± 2.6 (40)	45.8 ± 4.1† (40)
L2	8.4 ± 2.1 (42)	8.3 ± 1.8 (42)	15.4 ± 1.5 (42)	17.2 ± 1.7† (37)	6 ± 2 (42)	8 ± 4† (42)	6 ± 2 (42)	6 ± 2 (38)	45.0 ± 3.3 (42)	49.2 ± 4.5† (40)
L3	10.2 ± 2.5 (42)	10.0 ± 1.9 (38)	15.4 ± 1.6 (42)	16.9 ± 1.3† (38)	7 ± 3 (42)	13 ± 4† (42)	6 ± 2 (42)	7 ± 3 (38)	46.7 ± 3.5 (42)	51.7 ± 5.1† (40)
L4	13.2 ± 2.5 (42)	12.6 ± 2.3 (37)	15.4 ± 2.0 (40)	15.6 ± 1.5 (37)	11 ± 5 (42)	16 ± 5† (42)	7 ± 2 (42)	8 ± 3† (42)	47.3 ± 3.5 (42)	55.3 ± 5.4† (40)
L5	20.1 ± 3.7 (42)	16.6 ± 2.5† (39)	16.2 ± 2.1 (42)	13.8 ± 1.9† (39)	22 ± 5 (42)	29 ± 6† (42)	7 ± 4 (30) −9 ± 5‡ (12)	7 ± 4 (30) −10 ± 5‡ (12)	44.0 ± 3.9 (42)	49.2 ± 5.8†

* Data are given as means and standard deviations. The numbers in parentheses are the numbers of examples that were analyzed.
† Statistically significant compared with the value for the direct measurement (Student t test, $p \leq 0.05$).
‡ Sagittal angles at the fifth lumbar level occurred in either a cephalad or a caudad direction and are reported separately here.

TABLE III

MEASUREMENT OF LUMBAR PEDICLES ON RADIOGRAPHS AND COMPUTERIZED TOMOGRAPHIC SCANS OF LIVING YOUNG-ADULT PATIENTS*

	Transverse Diameter (mm)		Sagittal Diameter (mm)		Transverse Angle† (Degrees)		Sagittal Angle (Degrees)		Posterior-to-Anterior Distance† (mm)	
Level	M	F	M	F	M	F	M	F	M	F
L1	9.0 ± 1.9 (44)	6.3 ± 1.2‡ (54)	16.7 ± 1.9 (44)	15.4 ± 1.4‡ (54)	ND	ND	7 ± 2 (42)	6 ± 2 (46)	ND	ND
L2	9.9 ± 2.0 (46)	7.1 ± 1.3‡ (54)	16.8 ± 2.4 (46)	15.1 ± 1.5‡ (56)	9 ± 1 (2)	10 ± 1 (6)	6 ± 2 (44)	6 ± 2 (50)	53.3 ± 0.6 (2)	49.7 ± 4.7 (6)
L3	11.8 ± 2.0 (46)	8.2 ± 1.4‡ (54)	17.2 ± 2.6 (46)	15.0 ± 1.5‡ (55)	11 ± 2 (24)	12 ± 3 (43)	7 ± 2 (44)	6 ± 3 (50)	53.3 ± 3.2 (24)	47.0 ± 3.4‡ (41)
L4	13.7 ± 2.0 (46)	10.0 ± 1.7‡ (54)	16.3 ± 2.0 (45)	14.2 ± 1.7‡ (56)	13 ± 3 (34)	13 ± 4 (46)	7 ± 3 (44)	6 ± 3 (50)	55.4 ± 3.0 (34)	50.2 ± 4.7‡ (46)
L5	16.7 ± 2.2 (36)	13.5 ± 2.0‡ (46)	14.8 ± 1.7 (38)	13.2 ± 2.2‡ (50)	23 ± 8 (34)	23 ± 5 (46)	7 ± 3§ (42)	5 ± 5§ (50)	52.0 ± 4.5 (34)	46.0 ± 5.7‡ (46)

* Data are given as means and standard deviations. The numbers in parentheses are the numbers of examples that were analyzed.
† ND = no data.
‡ Statistically significant compared with the value for the men (Student t test, $p \leq 0.05$).
§ Caudally directed angles are reported in the text and are not recorded here.

The transverse angles of the pedicles of the first and second lumbar vertebrae were significantly smaller in the specimens from women. The largest transverse angles were in the fifth lumbar vertebra for both sexes, with a mean of 17 degrees for the specimens from men and 18 degrees for those from women.

For the first through fourth lumbar levels, the mean sagittal angles ranged in a cephalad direction from +5 to +7 degrees for both sexes, without statistically significant differences. Six pedicles were directed caudad and were not included in the means for these levels: one in the first lumbar vertebra in a man (−4 degrees), one in the second lumbar vertebra in a woman (−3 degrees), and four in the third lumbar vertebra (two in the same woman, at −4 and −5 degrees; one in another woman, at −12 degrees; and one in a man, at −8 degrees). In four vertebrae that had a caudad direction of the sagittal angle of the pedicle, the direction of the sagittal angle of the other pedicle of the same vertebra was cephalad.

The sagittal angles of the fifth lumbar vertebra for both sexes varied widely in both directions and were studied separately. The mean cephalad angle was +5 degrees for the men and +8 degrees for the women (this difference was statistically significant), and the mean caudad angle was −8 degrees for the men and −9 degrees for the women. The sagittal angles of both pedicles in each of the fifth lumbar vertebrae were in the same direction, whether cephalad or caudad.

The shortest distances from the posterior aspect of the laminar cortex to the anterior aspect of the cortex of the vertebral body along the central axis of the pedicle were found in the first lumbar vertebra for both sexes, with a mean of 45.0 millimeters for the men and 42.5 millimeters for the women.

The greatest distances for both sexes were found in the fourth lumbar vertebra, with a mean of 50.4 millimeters for the men and 45.9 millimeters for the women. These differences were statistically significant.

Part 2: Comparison of Radiographic Measurements of the Lumbar Pedicles with Direct Measurements (Table II)

The values that were obtained by measurements on radiographs of twenty-one of the forty-nine cadaver spines in Part 1 were compared with those obtained by direct measurement. For this purpose, the data from both sexes were combined.

The radiographic measurements of the transverse diameters of the fifth lumbar vertebra were smaller than the direct measurements, with statistical significance. Differences at the other vertebral levels were not significant.

The radiographic measurements of the sagittal diameters showed statistically significant differences compared with the direct measurements in the first, second, and third lumbar vertebrae, where they were larger than the direct measurements, and in the fifth lumbar vertebra, where they were smaller.

The radiographic measurements of the transverse angles were consistently larger than the direct measurements, with statistical significance, except in the first lumbar vertebra (not larger).

The radiographic and direct measurements of the sagittal angles were only slightly different at all five levels; these were calculated separately according to whether they were cephalad (positive) or caudad (negative). In five pedicles of the cephalad three lumbar vertebrae in both measurement groups, the angles were not included in the calculation of means for these levels because they were found to be caudad in direction: one at the first lumbar level (−4 degrees by direct and −5 degrees by radiographic measurement), one at the second lumbar level (−3 degrees by direct and −6 degrees by radiographic measurement), and three at the third lumbar level (two at −4 and −5 degrees by direct and −6 degrees by radiographic measurement and one at −8 degrees by both techniques). In three of the five vertebrae that had a caudad-directed pedicle, the sagittal angle of the other pedicle of the same vertebra was cephalad in direction. The sagittal angles of the pedicles

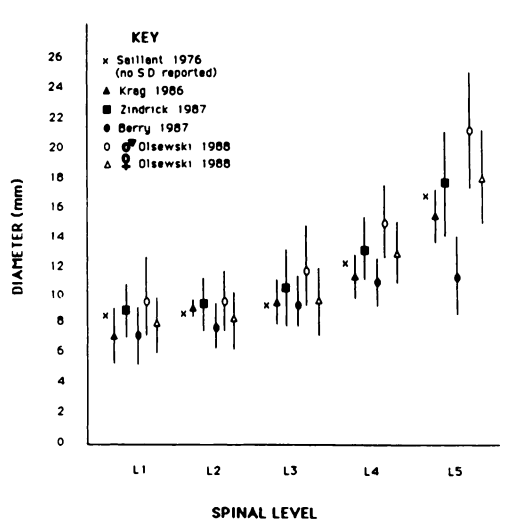


FIG. 7

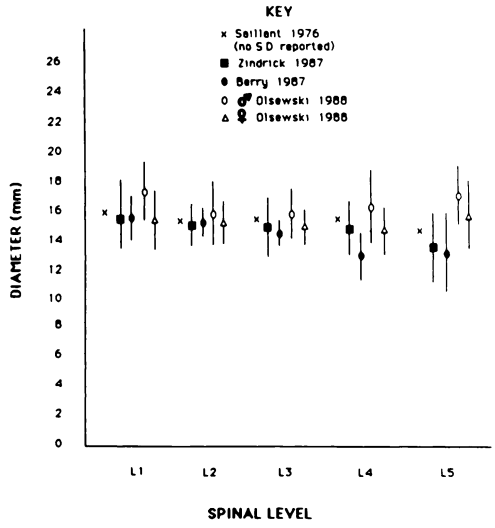


FIG. 8

Figs. 7 through 11: Graphs showing the direct measurements of lumbar vertebrae from cadavera compared with data from previously published studies. The points and lines represent means and standard deviations.
Fig. 7: Transverse diameters.
Fig. 8: Sagittal diameters.

of the fifth lumbar vertebra in both groups varied widely in both directions and were studied separately, and the means of the cephalad and caudad-directed angles also were calculated separately.

The radiographic measurements of the distance from the posterior aspect of the laminar cortex to the anterior aspect of the cortex of the vertebral body were consistently larger than the direct measurements, by 2.5 to 8.0 millimeters. This difference was found to be significant.

Part 3: Measurement of Lumbar Pedicles from Radiographs and Computerized Tomographic Scans of Living Young-Adult Patients (Table III)

For both the transverse and the sagittal diameters of the vertebrae, statistically significant differences were found between the sexes: the diameters were consistently smaller at each lumbar level in the women.

Because transverse angles were obtained retrospectively from routine computerized-tomographic scans of the lumbosacral spine, and because the scans for patients who had low-back symptoms did not usually include the first lumbar level, no data were available for that level. No statistically significant differences were found between the sexes at the remaining lumbar levels.

For all lumbar levels, the mean sagittal angles in a cephalad direction ranged from +5 to +7 degrees for both sexes, with no statistical differences. The sagittal angles of eight pedicles were not included in the calculation of means for these levels because they were caudad in direction: six at the first lumbar vertebral level (two at −3 degrees, two at −4 degrees, and two at −5 degrees, with each pair from the same patient) and two at the fifth lumbar level (both −15 degrees and in the same patient).

Because distances from the posterior aspect of the laminar cortex to the anterior aspect of the cortex of the vertebral body along the central axis of the pedicle were obtained from the same computerized tomographic scans, no data were available for the first lumbar vertebra. Statistically significant differences were found between the sexes for the third, fourth, and fifth lumbar vertebrae, with the distance being consistently smaller in the women.

Discussion

Values of dimensions and angles from direct measurements of cadavera (Part 1) compared with those from previous studies[7,26,52,63] are shown in Figures 7 through 11. There is good agreement for the stepwise increase in size from the first through fifth lumbar vertebra, but the mean transverse diameters for men in the current study were consistently larger than in the earlier studies, which combined data from both sexes. The mean sagittal diameters at each lumbar level for men in the present study also were consistently larger than those in other studies[7,52,63].

There is good agreement between our findings and those of Saillant in cadavera[52] except at the fifth lumbar vertebral level, where the present study showed a substantially smaller transverse angle. The present study showed substantially smaller transverse angles at all lumbar levels compared with previous investigations other than Saillant's[52].

Zindrick et al.[63] found an increased incidence of caudad-directed sagittal angles at all lumbar levels. Authors of previous studies of sagittal angles have calculated means from algebraic sums of angles in both the cephalad and the caudad direction for the fifth lumbar vertebra; in the present study, however, means were calculated separately for angles in each direction[52,63]. Use of an algebraic sum of the cephalad (positive) and caudad (negative) values of the sagittal angles in our study to calculate the mean sagittal angles yielded a value similar to those previously published. We

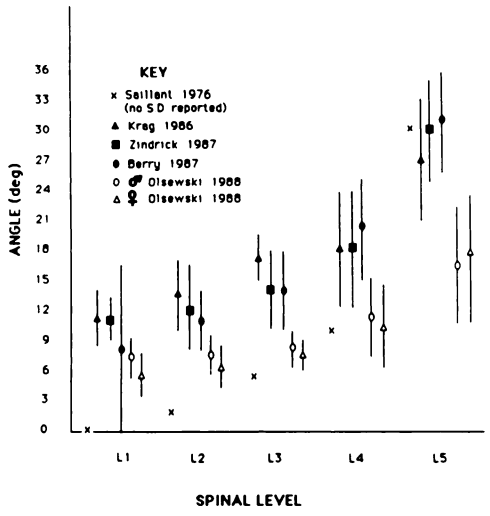


FIG. 9

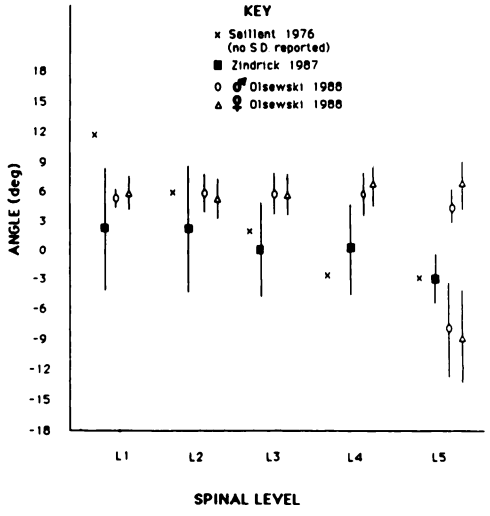


FIG. 10

Fig. 9: Transverse angles.
Fig. 10: Sagittal angles. The means for cephalad and caudad angles at the fifth lumbar level are given separately.

recommend, however, that for the fifth lumbar vertebra the mean cephalad and caudad-directed sagittal angles be calculated separately (Tables I and III), rather than from their algebraic sums, because averaging from an algebraic sum[52,63] gives a false indication of the actual angles in patients (Fig. 10).

The transverse and sagittal angles in our study were quite similar for the sexes (Tables I and III). The 1 to 2-degree difference between the sexes at the first and second lumbar levels (Table I) probably has no clinical importance. The lack of statistically significant differences between the sexes for these angles is consistent with the findings of van Schaik et al.[57]. Krag et al.[26], Zindrick et al.[63], and Berry et al.[7] all found the greatest distance from the posterior aspect of the laminar cortex to the anterior aspect of the cortex of the vertebral body in the second lumbar vertebra, but we found it in the fourth lumbar vertebra. We are in agreement with Krag et al. and with Berry et al. that this distance is substantially less in the fifth lumbar vertebra than in the fourth lumbar vertebra.

A major new observation in this study is the statistically significant difference in measurements according to the sex of the subjects. Comparative morphometric studies of the sexes, by both direct observation and radiographic analysis, revealed that dimensions are smaller in female subjects, but the transverse and sagittal angles do not differ much between the sexes. This finding appears to be consistent with the fact that men are, on the average, taller and heavier than women. Unfortunately, the heights and weights were not available for all specimens and patients in our study.

With the exception of those of the fifth lumbar vertebra, the pedicles are oval-shaped and narrowest in the transverse plane (Table I). The results of this study suggest that pedicles of lumbar vertebrae of men may safely accommodate substantially larger screws than has been previously reported[7,26,52,63] (Figs. 7 and 8). Whether it is beneficial to use screws with a larger diameter remains to be proved[26], but the present data should aid in establishing, for each sex, upper limits of the size of screws; above this limit, the pedicle may break or the screw may cut out[26,63].

Knowledge of the distance from the posterior aspect of the laminar cortex to the anterior aspect of the cortex of the vertebral body is important for the prevention of perforation of the anterior aspect of the cortex by screws and injury to vital structures[26,63]. For example, the difference in this distance between the sexes ranged from 2.5 to 5.0

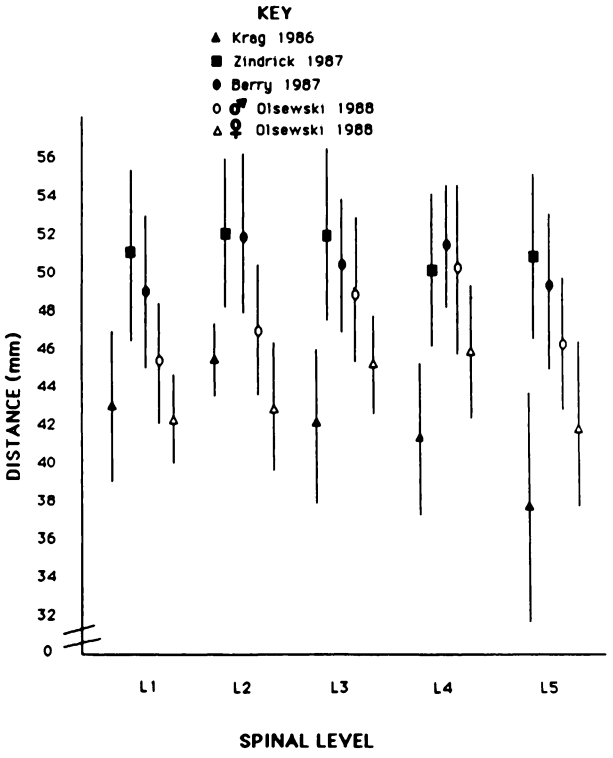


FIG. 11

The distance from the posterior aspect of the laminar cortex to the anterior aspect of the cortex of the vertebral body along the central axis of the pedicle.

millimeters (depending on the lumbar level), as directly measured in specimens from cadavera (Table I).

As important as sexual differences in the morphology of the pedicles is the observation that the diameters of pedicles measured on radiographs or computerized tomographic scans may differ significantly from the actual measurements of vertebrae from cadavera. Even when there is no detectable magnification, measurements of diameters on anteroposterior radiographs are different from those that are made directly (Table II). This is true at the fifth lumbar level even when the radiograph is of a disarticulated lumbar spine, and the image has much less overlap than that on a radiograph made *in vivo*. This observation may be related to previous findings[57] that suggested that the lateral borders of pedicles of the fifth lumbar vertebra are not clearly visible on anteroposterior radiographs.

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